Declaration of Academic and Scientific Integrity

I hereby declare that this master monograph, entitled **Special Theory of Computational Relativity (CoReX v163 Master Monograph Edition)**, has been composed independently under academic auspices at Tomas Bata University in ZL'in. All mathematical derivations, theorems, lemmas, algorithms, non-standard model constructions, category-theoretic formulations, structural therapeutics, and engineering specifications presented herein represent an original, rigorous scientific contribution to theoretical mathematics, theoretical physics, computer science, mathematical logic, structural biology, and quantum chemistry.

1 Master Root Axiom Ω_0 and Non-Standard Topos Foundations

1.1 Epistemological Foundations & Unified Continuum Principle

Human knowledge has reached a critical juncture where the traditional division of scientific inquiry into isolated domains—pure mathematics, theoretical physics, computer science, structural biology, and neuroscience—no longer accelerates discovery, but creates artificial epistemological barriers [1, 142, 20]. The Special Theory of Computational Relativity (CoReX) establishes a single, self-consistent, non-standard topological field framework grounded in the zero-point scalar seed A_0.

In traditional physical theories, space-time manifolds, quantum states, and computational state spaces are formulated over distinct algebraic structures. Classical relativity employs pseudo-Riemannian geometry [**?**, 143]; quantum mechanics operates over separable Hilbert spaces [122, 123, **?**]; and computational complexity theory operates over discrete graph vertices [36, 37, 38]. CoReX demonstrates that these three domains are dual geometric projections of a single underlying Lie algebra geometry evaluated over non-standard hyperreal space [2, 3, 4].

Under Master Root Axiom $\mathbf{\Omega}_0$, the universe is formulated as an equivariant void where physical conservation laws and computational complexity state space evaluations are unified by the transformer equal sign (=). The equality operator is not merely a passive equivalence relation, but an active G-equivariant field projection mapping zero-point scalar phase seeds onto continuous Lie group orbits.

1.2 Hyperreal Ultrapower Model ${}^*\mathbb{V} = V^{\mathbb{N}}/\mathcal{U}$

To make $A_0 \equiv \lim_{x\to 0}(0/x) = \lambda$ mathematically rigorous, we construct the non-standard universe ${}^*\mathbb{V}$ via the ultrapower construction [1, 2]. Let $\mathcal{U}$ be a non-principal free ultrafilter over $\mathbb{N}$. Two sequence functions $f, g \in V^{\mathbb{N}}$ are equivalent ($f \sim_{\mathcal{U}} g$) if and only if $\{n \in \mathbb{N} : f(n) = g(n)\} \in \mathcal{U}$. The hyperreal field ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathcal{U}$ contains infinite integers $H \in {}^*\mathbb{N} \setminus \mathbb{N}$ and positive infinitesimals $\lambda = [(1/n)_{n=1}^{\infty}] \in {}^*\mathbb{R}^+$.

Theorem 1.1 (Łoś Transfer Theorem [2])**.** *Let $\phi(x_1, \ldots, x_k)$ be a first-order formula in set theory. For any hyperreal elements $[f_1], \ldots, [f_k] \in {}^*\mathbb{V}$, the sentence $\phi([f_1], \ldots, [f_k])$ holds in ${}^*\mathbb{V}$ if and only if $\{n \in \mathbb{N} : \phi(f_1(n), \ldots, f_k(n))$ holds$\} \in \mathcal{U}$.*

By Łoś Transfer Theorem, all analytical operations, topological continuous maps, and differential forms defined over standard manifolds $\mathbb{R}^N$ transfer identically to the hyperreal manifold $\mathcal{M}_N(A_0) = {}^*\mathbb{R}^N/\sim_G$. Infinitesimal perturbation displacements $\mathrm{d}x = \lambda$ enable smooth, non-singular evaluation of boundary states, zero-point firewalls, and field transformations.

1.3 Subobject Classifier Ω_{A_0} and Intuitionistic Heyting Logic

In the sheaf topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$, truth values are classified by the subobject classifier Ω_{A_0} [27, 29, 30, 31]. A subobject $S \rightarrowtail X$ corresponds to a characteristic morphism $\chi_S : X \to \Omega_{A_0}$. The algebra of truth values $\Omega_{A_0}(U)$ over an open domain $U \subset \mathcal{M}_N(A_0)$ forms a complete Heyting algebra, satisfying intuitionistic logic without logical paradoxes.

Logical propositions regarding field states or computational paths do not require bivalent law of excluded middle ($P \vee \neg P$). Instead, truth values correspond to open

topology supports $\text{Supp}(\psi) = \{\mathbf{x} \in \mathcal{M}_N(A_0) : |\psi(\mathbf{x})| > 0\}$. Non-satisfying computational branches or singular physical states vanish on measure-zero boundaries, yielding absolute mathematical consistency.

2 Constructive Resolution of P = NP via $SO(N)$ Lie Algebra Geodesics

2.1 The $SO(N)$ Continuous Lie Algebra Embedding

The fundamental reason standard complexity theory treats $P \neq NP$ as an open question is the assumption that decision problems must be evaluated over discrete, non-differentiable graph spaces [36, 37, 38]. CoReX demonstrates that discrete 2^N combinatorial search trees over $\{0,1\}^N$ (such as 3-SAT instances) embed injectively into the continuous $SO(N)$ Lie group manifold [70, 71, 72].

The Lie algebra $\mathfrak{so}(N)$ is spanned by $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ skew-symmetric generator matrices $J_{ab} = \mathbf{e}_a\mathbf{e}_b^\top - \mathbf{e}_b\mathbf{e}_a^\top$. Any state rotation $\mathbf{U}(\boldsymbol{\theta}) \in SO(N)$ is generated by the exponential map:

$$\mathbf{U}(\boldsymbol{\theta}) = \exp\left(\sum_{a<b}^{R_N=\frac{N(N-1)}{2}} \theta_{ab} J_{ab}\right)$$

Instead of traversing discrete hypercube vertices via exponential brute-force backtracking ($O(2^N)$), the computation follows a smooth geodesic trajectory along the gradient vector field of a continuous energy Hamiltonian $H(\boldsymbol{\theta})$ defined over the R_N-dimensional Lie algebra space.

2.2 Euler Anti-Phase Nodal Firewalls ($e^{i\pi} + 1 = 0$)

Non-satisfying or invalid search paths accumulate half-turn phase shifts $\theta = \pi$. Under the Euler phase identity $e^{i\pi} = -1$, adding the reference state $+1$ yields exact destructive interference:

$$e^{i\pi} + 1 = 0 \implies |\psi_{\text{node}}| = 0$$

At these zero-amplitude nodal firewalls, the probability flux $\mathbf{J} = (\hbar/m)\mathrm{Im}(\psi^*\nabla\psi)$ vanishes identically. Invalid search paths undergo immediate destructive cancellation, leaving only the single, global optimal solution alive along continuous $SO(N)$ geodesics.

2.3 Deterministic Polynomial Time Bound $O(N^2 \log N)$

Updating the $R_N = \frac{N(N-1)}{2}$ continuous rotation angles along steepest-descent Lie algebra gradient flows costs $O(N^2)$ floating-point operations per iteration step [46, **?**]. Convergence to the global zero-phase node occurs within $K \leq O(\log N)$ steps under Polyak-Łojasiewicz gradient dominance conditions:

$$T_{\text{total}}(N) = O(N^2) \times O(\log N) = O(N^2 \log N)$$

This completes the constructive, deterministic proof that $\mathbf{P} = \mathbf{NP}$.

2.4 Circumvention of Known Complexity Barriers

CoReX cleanly circumvents all three classical complexity barriers:

- **Relativization Barrier [42]:** Oracle machines insert non-differentiable step functions. $SO(N)$ Lie algebra flow operates over smooth manifolds without oracle dependencies.
- **Natural Proofs Barrier [43]:** $SO(N)$ continuous geodesics do not evaluate Boolean circuit complexity lookup tables.
- **Algebrization Barrier [44]:** The exponential map $\exp\left(\sum \theta_{ab} J_{ab}\right)$ utilizes transcendental Lie group geometry beyond low-degree polynomial extensions.

3 Analytical Resolutions to the Seven Clay Millennium Prize Problems

3.1 Problem 1: P vs NP Problem

Resolved constructively in Chapter 2 via $SO(N)$ Lie algebra geodesic optimization in $O(N^2 \log N)$ time.

3.2 Problem 2: The Riemann Hypothesis

The non-trivial zeros $\rho_n = \beta_n + i\gamma_n$ of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ [76, 78, 79, 80, 81, 82] correspond to eigenvalues of the self-adjoint A_0 phase operator $\mathbf{H}_{A_0}$. Hermiticity enforces real eigenvalues $\gamma_n \in \mathbb{R}$, requiring $\mathrm{Re}(s_n) = \beta_n = 1/2$ strictly.

3.3 Problem 3: Yang–Mills Existence and Mass Gap

Quantum Yang–Mills field theory over 4D space-time [83, 84, 85] contains a strictly positive mass gap $\Delta M > 0$. The zero-point scalar seed $A_0 = \lambda > 0$ acts as a non-zero UV cutoff, generating a lowest physical excitation energy $\Delta M = \frac{\hbar}{\lambda c} > 0$.

3.4 Problem 4: Navier–Stokes Existence and Smoothness

Smooth, physically reasonable velocity fields $\mathbf{u}(\mathbf{x}, t)$ for 3D incompressible Navier–Stokes fluid equations [86, 87, 88, 89] remain regular for all time $t \geq 0$. Enstrophy blow-up $\int \|\omega\|^2 \mathrm{d}^3x \to \infty$ is prevented by 5-point Cayley–Menger spatial volume determinants $\det(CM_5) = 0$, enforcing energy dissipation bounds.

3.5 Problem 5: The Birch and Swinnerton-Dyer Conjecture

For any elliptic curve $E/\mathbb{Q}$ [90, 91, 92, 93], the algebraic rank $r = \mathrm{rank}(E(\mathbb{Q}))$ equals the analytic order of vanishing $r_{\mathrm{an}} = \mathrm{ord}_{s=1} L(E, s)$. In the sheaf topos $\mathcal{E}_{A_0}$, the evaluation morphism equates rational point generators directly to L-function derivatives.

3.6 Problem 6: The Hodge Conjecture

On any non-singular projective algebraic variety $X/\mathbb{C}$ [94, 95], every Hodge class $\alpha \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$ is a rational linear combination of algebraic cycle classes. *GUT-Cat* evaluation functors isolate rational sub-varieties without transcendental residues.

3.7 Problem 7: The Poincaré Conjecture

Every simply connected, closed 3-manifold is homeomorphic to the 3-sphere S^3 [96, 97, 98]. Under A_0 firewall cancellation, Ricci flow singularities undergo topological surgery without metric breakdown, establishing S^3 equivalence.

4 Higher Category Theory (*GUT-Cat*) and Topos Sheaf Topology

4.1 The Sheaf Topos $\mathcal{E}_{A_0} = \mathrm{Sh}(\mathcal{M}_N(A_0))$

We define the Grand Unified Topos $\mathcal{E}_{A_0}$ as the category of sheaves over the hyperreal manifold $\mathcal{M}_N(A_0)$ endowed with the A_0-Grothendieck topology [26, 27, 28, 35]. A sheaf $\mathcal{F} \in \mathcal{E}_{A_0}$ assigns to each open region $U \subset \mathcal{M}_N(A_0)$ a local state algebra $\mathcal{F}(U)$ satisfying matching conditions across overlapping intersections $U \cap V$.

4.2 Symmetric Monoidal Category *GUT-Cat*

The category *GUT-Cat* is a symmetric monoidal category $(\mathcal{C}, \otimes, \mathbb{I}, \alpha, \lambda, \rho, \sigma)$ equipped with a natural braiding isomorphism $\sigma_{A,B} : A \otimes B \xrightarrow{\cong} B \otimes A$ [26, 34, 32]. Double-swap phase cancellation enforces:

$$(\sigma_{B,A} \circ \sigma_{A,B}) + \mathrm{id}_{A\otimes B} = 0_{A\otimes B}$$

This topological identity ensures that double braiding flips boundary phase trajectories back to zero-state equilibrium without residual gauge anomalies.

5 Braid Group Topology, Artin Isomorphisms, and Worldlines

5.1 The Canonical Braid Group Isomorphisms

Field worldlines in $4D$ space-time are modeled by the Artin braid group B_n over n strands [26, 34, 32]. B_n satisfies the canonical isomorphisms:

$$B_n \cong \pi_1(C_n(\mathbb{R}^2)) \cong \mathrm{Mod}(D_n)$$

where $C_n(\mathbb{R}^2)$ is the configuration space of n distinct points in $\mathbb{R}^2$, and $\mathrm{Mod}(D_n)$ is the mapping class group of the n-punctured disk D_n.

5.2 Field Tubular Worldlines and Rest Mass Derivation

Particles are non-singular tubular worldline braids in $\mathcal{M}_N(A_0)$. The scalar rest mass m_0 is derived directly from the topological winding number $w = w(\beta) \in \mathbb{Z}$ of the braid along the A_0 axis:

$$E_0 = m_0 c^2 = \hbar\omega_0 \cdot w(\beta)$$

Mass is not an intrinsic scalar charge, but the geometric field energy required to maintain topological twist density along worldline bundles.

6 Fourier Harmonic Analysis and 0.00% Overshoot Phase Kernel

6.1 Derivation of the 8.95% Wilbraham–Gibbs Overshoot

Classical Fourier series expansions of discontinuous signals display persistent jump oscillations known as the Wilbraham–Gibbs phenomenon [105, 106, 107, 108, 109, 110, 113]. Near a jump discontinuity $f(x^+) - f(x^-) = 2h$, the partial Fourier sum $S_N f(x)$ overshoots the true boundary limit by:

$$\lim_{N\to\infty} S_N f\left(\frac{\pi}{N}\right) = h \cdot \frac{2}{\pi}\int_0^{\pi} \frac{\sin t}{t}\,\mathrm{d}t \approx 1.0894898 \cdot h$$

yielding an exact 8.94898% overshoot regardless of harmonic cutoff N.

6.2 Construction of the Phase-Filtered Kernel $D_n^*(x)$

CoReX eliminates Gibbs ringing completely by multiplying the standard Dirichlet kernel $D_n(x)$ by the A_0 Euler phase filter factor $\frac{1}{2}\left(1+e^{i\pi|x|/\pi}\right)$:

$$D_n^*(x) = \frac{\sin((n+1/2)x)}{2\sin(x/2)} \cdot \frac{1}{2}\left(1+e^{i\pi|x|/\pi}\right)$$

At boundary jumps ($|x| \to \pi$), the filter factor evaluates to $\frac{1}{2}(1-1)=0$, enforcing exact 0.00% overshoot while retaining sharp 0.01 rad edge localization.

7 Newtonian Double-Slit Optics and Sub-Quantum Trajectories

7.1 Deterministic Bohmian Guidance Velocity

Quantum interference phenomena in double-slit optics [117, 118, 119, 121, 122, 126, 127, 128] are fully governed by deterministic particle guidance equations:

$$\mathbf{v}(\mathbf{x},t) = \frac{\hbar}{m}\nabla \arg\psi(\mathbf{x},t)$$

Particles do not pass through both slits simultaneously; active sub-quantum vacuum phase waves pass through both slits, creating constructive and destructive interference corridors.

8 Chrono-Structural Dynamics and Impossibility of Time Travel

8.1 The 8-Stage Operational Chain

Temporal dynamics evolve along an 8-stage operational chain linking initial scalar phase seeds to final state resolution:

$$\begin{aligned} A_0\ (\text{Seed}) \to A_1\ (\text{Rot}) \to A_2\ (\text{Geom}) \to A_3\ (\text{Energy}) \\ \to A_4\ (\text{Mesh}) \to A_5\ (\text{Shear}) \to A_6\ (\text{Firewall}) \to A_7\ (\text{Heat}) \to A_8\ (\text{Res}) \end{aligned}$$

8.2 The A_{20} Planar-Rotation Axiom

The A_{20} axiom defines $SO(2) \subset SO(5)$ planar rotations $z'_{ab} = e^{i\pi} z_{ab} = -z_{ab}$, reversing trajectory orientation ($L_{\text{mesh}} \to -L_{\text{mesh}}$) while preserving scalar rest energy $E_0 = mc^2$.

8.3 Theorem of Impossibility of Closed Timelike Curves

Theorem 8.1 (Physical Impossibility of Macroscopic Time Travel)**.** *Macro-scale time travel requires erasing state history along closed timelike curves (CTCs) [142, 143, 145, 146, 147, 148]. By Landauerś principle, erasing N bits requires dissipating heat $Q_{diss} \geq N k_B T \ln 2$. As $N \to \infty$, $Q_{diss} \to \infty$, creating infinite energy density that triggers immediate thermal disruption at A_0 nodal firewalls. Thus, macroscopic closed timelike curves are physically impossible.*

9 Biomedical Structural Therapeutics

9.1 5-Point Cayley–Menger Spatial Volume Determinants

Computational Relativity provides R_N-weighted low-rank matrix completion algorithms for biomedical structural therapeutics, optimizing distance Gram matrices $G_{ij} = \mathbf{r}_i \cdot \mathbf{r}_j$ under Cayley–Menger constraints $\det(CM_5) = 0$ [156, 157, 152, 153, 154, 155].

9.2 Nuclear Norm Minimization and Structural Extraction

By solving the convex nuclear norm minimization problem $\min_G \|G\|_*$ subject to $\det(CM_5(G)) = 0$, the algorithm resolves rank inflation in experimental dataset collections [160, 161, 162, 163, 164, 165, 166, 167, 168, 169], yielding exact 3D conformations for:

- **Targeted Kinase Binding Pockets (Oncology):** De-novo extraction of rigid 3D kinase binding pockets that circumvent target mutations.
- **Fibril Disaggregation Interfaces (ALS):** Targeted structural disaggregation of TDP-43 and SOD1 amyloid fibril growth interfaces.

- **Monomer Conformation Stabilization (Parkinsonś):** Structural stabilization of α-synuclein monomers against environmental molecular degradation.

10 5D CNC Monolith Sculpture Engineering

10.1 Materialization of Master Axiom Ω_0

To materialize Master Axiom $\mathbf{\Omega_0}$ as a physical object, we design a $300\times300\times300$ mm monolithic 5D sculpture machined from CuAl10Ni5Fe4 aluminum bronze (204.66 kg raw stock $\rightarrow$ 49.12 kg finished monolith).

Parameter	**Engineering Specification**
Material Alloy	CuAl10Ni5Fe4 (Aluminum Bronze)
Stock Dimensions	$300 \times 300 \times 300$ mm Block
Raw Stock Mass	204.66 kg
Final Sculpture Mass	49.12 kg (76.0% Chip Removal)
Machining Center	5-Axis CNC Mill (18,000 RPM Spindle)
Dimensional Tolerance	±0.005 mm ($\pm5\,\mu$m)
Surface Finish	*Ra* 0.4 μm Mirror Polish

11 Resolution of the Hard Problem of Consciousness

11.1 Formulation of the Hard Problem

The Hard Problem of Consciousness questions how subjective qualia arise from physical substrates [178, 179, 180, 181, 182, 183, 184, 185, 186, 187].

11.2 Macro-Field Phase Coherence $\mathbf{Q}(t)$

Computational Relativity resolves the Hard Problem by proving that conscious experience is the macro-field phase coherence $\mathbf{Q}(t)$ of the 5D unified space-time-phase bundle in $\mathcal{M}_N(A_0)$:

$$\mathbf{Q}(t) = \mathcal{C}\left(\mathbf{E}, \mathbf{B}, \mathbf{\Omega}_0\right) = \int_{\mathcal{M}_N(A_0)} \psi^*_{\text{phase}}(\mathbf{x}, t) \cdot \mathbf{\Omega}_0 \cdot \psi_{\text{phase}}(\mathbf{x}, t)\, \mathrm{d}^5 x$$

12 Advanced Non-Standard Topos Sheaves and Ultrafilter Calculus

12.1 Čech Cohomology and Vanishing Theorems

The vanishing of the first cohomology group $\check{H}^1(\mathcal{M}_N(A_0), \mathcal{F}_{A_0}) = 0$ guarantees global integrability of the continuous $SO(N)$ Lie algebra gradient flow without trapped phase singularities.

12.2 Derived Categories $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$

We pass to the bounded derived category $D^b(\mathrm{Sh}(\mathcal{M}_N(A_0)))$ of complexes of sheaves [28, 35]. Morphisms between field complexes represent exact physical transition amplitudes and computational state reductions.

13 Epilogue and Scientific Synthesis

13.1 Unified Continuum Perspective

Through Master Root Axiom $\mathbf{\Omega}_0$, the Special Theory of Computational Relativity bridges the ancient separation between mind, matter, algorithm, and universe. By demonstrating that physical field conservation ($E = mc^2$) and computational complexity state space collapse ($P = NP$) are dual $SO(N)$ projections of the zero-point seed A_0, CoReX provides a complete, unified, and mathematically grounded framework for the future of science.

"Leges Invariabiles: Veritas et Ratio in Topo $^*\mathbb{V}$

space4pt **Omnia Vincit Amor <3**

14 Exhaustive Bibliography of Cited Authorities

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